

PARTICULAR SPECIFICATION P10

Plant Performance Testing

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P10.1 PLANT PERFORMANCE TESTING

P10.1.1 Overview

- (a) Plant Performance to be demonstrated as part of Testing and Commissioning.
- (b) The final stage of testing and commissioning is Plant Performance Testing, the general objectives and overall procedures of which are described in General Specification G2.26 Testing and Commissioning.
- (c) This section of the Specifications describes the required operating regimes and durations to be adopted during the Plant Performance Test.

P10.1.2 Plant Performance Test Periods (PPTP)

- (a) There shall be qualified process engineers on-site with relevant THP and sidestream nitrogen removal process experience, in addition to the T&C team specified in General Specification G2.26 Testing and Commissioning.
- (b) The PPTP cannot reasonably take place until the overall commissioning of the Biosolids and Digesters or as directed by the S.O..
- (c) There shall be one common PPTP for the Digestion Pre-treatment facility and Sidestream Nitrogen Removal facility, which shall last for 28 consecutive days, during which the facility and equipment shall operate on a 24 hr, 7-day basis (24/7) to demonstrate throughput and specified performance.
- (d) For Digestion Pre-treatment facility, this total 28-day period shall include:
 - (i) Operating at Maximum Month flows and stress condition (Maximum THP unit hydraulic capacities) continuously for up to 7 days.
- (e) For Sidestream Nitrogen Removal facility, this total 28-day period shall include:
 - (i) Operating at Maximum Month flows and stress condition (Maximum Solids Separation and Deammonification Reactors hydraulic capacities) continuously for up to 7 days.
- (f) If the amount of feed is insufficient to allow the PPTP to be carried out at the hydraulic conditions as specified, the S.O. may at his absolute discretion, agree with the Contractor the sustainable feed flow to be used during the Plant Performance Test Period (PPTP) and the appropriate number of process units which can be reasonably tested under the prevailing flow conditions. Under such circumstances, the Contractor shall work with the S.O. to develop an appropriate PPTP methodology and evaluation criteria required to demonstrate the equipment performance guarantees with the plant operating at the design conditions and for suitable durations. Successful completion of the Performance Tests will be accepted at the S.O.'s discretion upon meeting the plant performance guarantees with the agreed maximum number of units operating. The number of passed days shall not exceed twenty-eight (28).

P10.1.3 Performance Requirement during PPTP

- (a) During the PPTP, the facilities shall operate entirely in auto mode without interruption, with guaranteed output of all process systems and equipment. The performance and reliability of the facilities shall be demonstrated during the PPTP.
- (b) During PPTP the Contactor shall prove and verify the performance of the Works to meet the Annex O – Schedule of Performance Guarantees and the Annex N – Schedule of Technical Data in Volume 1 as well as the requirements in the Specifications. In addition, the Contractor shall prove and verify the satisfactory operation of all components of the Works as follows:
 - (i) Digestion Pre-treatment
 - 1) Provide thermal hydrolysis of pre-dewatered sludge as specified;
 - 2) Provide Hydrolysed Sludge Cooling to maintain digester temperature;
 - 3) Heat recovery for hot water supply as specified;
 - 4) Provide Instrument Air supply to Biosolids Building consumers as specified; and
 - 5) Feeding of Hydrolysed Sludge to digesters.
 - (ii) Sidestream Nitrogen Removal
 - 1) Separation of solids before Deammonification process; and
 - 2) Provide Partial Nitritation and Anammox (Deammonification) process for Ammonia removal from Dewatering Centrate as specified.
 - (iii) Steam and Energy Consumption
 - 1) Meet the specific energy consumptions as specified in Annex O; and
 - 2) Meet the specific steam consumptions as specified in Annex O.
 - (iv) Quality Requirement
 - 1) Collection of samples at autosamplers and at selected sampling points including but not limited to those specified below to external accredited laboratory:
 - a) Total suspended solids, $\text{NH}_4\text{-N}$ for Deammonification reactor feed; and
 - b) $\text{NH}_4\text{-N}$ for Deammonification reactor effluent.

- 2) Additional Sampling as instructed by the S.O./the Board: Internal laboratory testing at any other sampling location as instructed by the Board.
- (c) On successful completion of the PPTP, the S.O. will issue a Certificate of Substantial Completion for the whole of Works.

P10.1.4 Testing, Sampling and Analysis Methods

- (a) The Contractor shall note that the laboratory within TWRP will not be ready during PPTP. If the Contractor intends to carry out the tests at the Site, it shall identify a suitable location within the Site to set up a temporary laboratory. The Contractor shall indicate its intention, meeting the following requirements, in the Testing and Commissioning Plan, subject to the S.O. for approval.
 - (i) The temporary laboratory and the equipment used in the temporary laboratory shall be set up, maintained and calibrated in accordance to ISO/IEC 17025 General Requirements for the Competence of Testing And Calibration Laboratories; and
 - (ii) The experienced water chemist responsible for overseeing the temporary laboratory operations shall have at least 5 years of experience working in a SAC-SINGLAS accredited laboratory and shall be at site as required during PPTP.
- (b) The Contractor shall provide manpower to assist the Board in the collection of samples for testing at its own laboratories during the PPTP. The samples shall be dropped off at one of the access points or other locations within TWRP as instructed by the S.O. to be pick-up.

P10.1.5 Test Failures

- (a) Failure to meet the above criteria on any day of 28 days PPTP shall constitute a “failed day”. One or more failure occurring on the same day shall constitute a single “failed day”.
- (b) One additional day shall be added to the 28-day PPTP for each “failed day”.
- (c) The final three (3) days of the PPTP shall be totally successful, consecutive days of uninterrupted operation. A failure of any of the days between 26 to 28 shall restart the PPTP at day 26 until three (3) consecutive, fully successful days have been achieved.

P10.1.6 Test Reports Submission

- (a) During the PPTP, the Contractor shall report daily, the recorded flow and quality parameters, and all failures to S.O. The Contractor shall also give reasons for the failures in the report.

- (b) The Contractor shall be responsible for collecting and collating all data and test results and shall carry out all necessary calculations to confirm compliance with the project Specifications. A report comprising test data, results, calculations and conclusions shall be submitted within 1 week of the date of completion of the respective plant performance tests, or set of tests.
- (c) If the report is not acceptable it shall be returned to the Contractor for verifications or amendments and resubmission. The Contractor shall submit three (3) bound copies of the reports within two (2) months of acceptance of the approved draft reports.

END OF SECTION